

Deep Analog Null Steering: Real-Time DoA Estimation for GNSS Multi-Jammer Mitigation on SWaP-Constrained Analog CRPAs

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Abstract—A controlled-reception-pattern antenna (CRPA) built on an *analog* beamforming network is attractive under size, weight and power constraints precisely because it carries one receiver chain instead of N — and for the same reason it cannot observe its own spatial covariance matrix. Everything the processor ever sees is a short vector of non-negative scalar power readings, one per applied probe weight: no phase, no in-phase/quadrature samples, no per-element data. This paper asks how much direction-of-arrival (DoA) information survives that measurement, and answers it for a four-element analog CRPA with full-wave simulated element patterns driven by an eight-probe codebook whose beams are *nulls* rather than mainlobes, so that the informative probe is the one reporting the *least* power. We first establish an observability ceiling that is a property of the antenna rather than of any algorithm: each probe power is a linear functional of the rank-structured covariance, so probe readings span at most 16 real dimensions no matter how many probes are read, capping the number of jointly identifiable jammers at 5; the familiar common-element-pattern idealisation collapses this to 9 dimensions and 2 jammers, which we show to be an artifact of the idealisation. Within that ceiling we train a 52,575-parameter hybrid estimator — a geometric anchor over subsector centres composed with an attention refiner — that reads the eight scalars directly. Against a single jammer it attains a median great-circle error of 3.717° and an RMSE of 7.072° , and the resulting projection null delivers a median jammer suppression ratio of 27.71 dB with $P(\text{JSR} \geq 20 \text{ dB}) = 89.3\%$. Statistical efficiency is strongly geometry-dependent — RMSE/CRB of $1.93\times$ on well-conditioned geometries and $0.70\times$ on ill-conditioned ones — so we report no single efficiency number, and we do not claim to beat the bound. A fixed-direction experiment over 1,008 directions separates bias from variance exactly and shows the sub-bound behaviour to be bias, not superefficiency: between the two conditioning halves the bias magnitude grows $4.5\times$ while the scatter moves by 7%, and the bias vectors point at the very subsector centres the anchor is built from. Finally, the multi-jammer limit is shown to be the array's and not the network's: from one to three jammers the scaled Fisher conditioning degrades $44.4\times$ and the bound itself degrades $8.63\times$, while the measured error rises only $3.02\times$.

Index Terms—GNSS, controlled-reception-pattern antenna, analog beamforming, jamming mitigation, direction of arrival, null steering, Cramér–Rao bound, identifiability, attention networks.

I. INTRODUCTION

GLOBAL navigation satellite signals arrive at the Earth's surface tens of decibels below the thermal noise floor, which makes a receiver trivially deniable: a low-power emitter anywhere in the sky can raise the interference level past

the point where the correlators lock. The standard spatial answer is a controlled-reception-pattern antenna, an array whose combining weights are adapted so that the pattern carries a null toward each interferer. In a *digital* CRPA this is routine, because every element has its own receiver chain and the processor holds the full snapshot record; the sample covariance matrix is directly available and half a century of adaptive array theory applies to it without modification [1]–[3].

That architecture is, however, the single most expensive part of the receiver. N elements require N down-conversion chains, N analog-to-digital converters and the clock distribution to keep them coherent — exactly the cost, mass and power that a small platform does not have. The size-weight-and-power-constrained alternative is to move the combining in front of the receiver: a single analog beamforming network applies a programmable complex weight to each element, sums the result in the RF domain, and presents *one* chain to *one* converter. The saving is real and the loss is severe. Applying a weight vector $\mathbf{w}_m$ and reading the resulting power yields a single non-negative scalar. Phase is gone. The per-element signals are gone. The covariance matrix, which is the object every classical adaptive algorithm consumes, is never formed and cannot be formed by inspection. What the processor holds instead, after a scan of M weights, is a vector $\mathbf{p} \in \mathbb{R}_+^M$ of powers.

This paper takes that measurement literally and asks what direction-finding performance is attainable from it. The array under study has four elements on a 6 cm square — a circumference of 0.224λ at 1580 MHz, so the aperture is small even by CRPA standards — and its element patterns come from a full-wave simulation of the mounted array rather than from an isotropic idealisation. Its probe codebook has a feature that turns out to organise the entire problem: the probes are *nulling* beams. Probe m is designed to place a deep null on the centre of subsector m , so a jammer sitting near that centre makes probe m report the *smallest* power in the scan, not the largest. The measurement is therefore read by an argmin, not an argmax, and the useful signal lives in the depth and shape of a notch. On our codebook the on-centre null is 90.49 dB deep at the median, and a raw argmin over the eight probe readings identifies the occupied subsector exactly 70.0% of the time against a chance rate of 12.5%. The information is unmistakably present; the question is how much of it can be extracted, and what stops it from being all of it.

The first contribution of the paper is that the ceiling is set by the antenna and not by the estimator. The mean power returned

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by probe m is a *linear* functional of the spatial covariance $\mathbf{R}$, and $\mathbf{R}$ for four elements is a Hermitian matrix carrying $4^2 = 16$ real degrees of freedom. Consequently the entire family of probe measurements — every probe that could ever be designed, in any number — spans at most 16 real dimensions. Reading more probes buys averaging, never new dimensions. Since a scene of J incoherent jammers plus noise is described by $3J+1$ real parameters, joint identifiability requires $J \leq 5$, and at $J = 3$ ten of the 16 available dimensions are already spent. We also revisit the widely quoted common-element-pattern count, which collapses the accessible span to 9 dimensions and would cap J at 2: that collapse is a degeneracy of the idealisation, and the full-wave pattern diversity of a real mounted array lifts it.

The second contribution is an estimator matched to the measurement rather than retrofitted to it. Because the probes are nulls, the natural first-order readout is a soft argmin over subsector centres; because a soft argmin over eight fixed centres can only ever return a point inside their convex hull, it must be composed with a refiner that can leave that hull. We therefore use a small hybrid network — 52,575 parameters in total — consisting of a geometric anchor that forms a convex combination of the eight subsector-centre encodings, and a multi-head attention refiner that reads the probe vector as a short token sequence and emits a correction. The correction is parameterised so that the returned direction lies in the upper hemisphere *by construction*: the elevation correction is applied in the logit of $\cos \theta$ and the azimuth as a unit-circle residual, so no folding, clipping or wrapping occurs anywhere in the model. Inference is a single forward pass; at deployment the latency is dominated by the probe scan itself, not by the network.

The third contribution is a diagnostic one, and it is the reason the paper reports what it does. Measured against a per-scene Cramér–Rao bound the estimator’s error ratio is above unity on well-conditioned geometries and *below* unity on ill-conditioned ones. Rather than quote the aggregate number — which would amount to advertising a bound violation — we designed a fixed-direction experiment that separates bias from scatter exactly, by repeating independent measurements at each of 1,008 fixed directions. The sub-bound cells are bias-dominated, the bias grows by a factor of 4.5 between the conditioning halves while the scatter barely moves, and the bias vectors point inward toward the subsector centres from which the anchor is built. The sub-bound behaviour is thus a property of a biased estimator, for which the unbiased bound is simply not the right yardstick, and nowhere in this paper do we claim to have beaten it.

The remainder of the introduction fixes the array, the sectorisation, the probe convention, the measurement statistics and the estimation problem itself, in that order, because several later results are unreadable without the conventions stated precisely.

A. System Model and Measurement

1) Array and Full-Wave Simulated Element Patterns

The antenna is a four-element planar array, $N = 4$, with the elements at the corners of a square of side 6 cm in the $z = 0$ plane,

$$\mathbf{p}_n \in \{(\pm 0.03, \pm 0.03, 0)\} \text{ m}, \quad n = 1, \dots, 4. \quad (1)$$

The circumradius is $0.03\sqrt{2} = 42.4$ mm, which at the design frequency $f_0 = 1580$ MHz ($\lambda = 189.7$ mm) is 0.224λ . The aperture is small, and this single number governs much of what follows: angular resolution, the conditioning of the multi-jammer problem and the width of an achievable null are all aperture-limited.

Directions are given in the upper hemisphere by elevation-from-zenith $\theta \in [0^\circ, 90^\circ]$ and azimuth $\phi \in [0^\circ, 360^\circ)$, with wave vector

$$\mathbf{k}(\theta, \phi) = \frac{2\pi}{\lambda} [\sin \theta \cos \phi \quad \sin \theta \sin \phi \quad \cos \theta]^\top. \quad (2)$$

The array response is not the ideal phase-only manifold. Each element carries its own complex pattern $E_n(\theta, \phi)$, obtained from a full-wave simulation of the mounted array on a 1° grid in both angles and evaluated at arbitrary directions by bilinear interpolation, so that mutual coupling and the platform are represented rather than idealised away. The effective steering vector is therefore

$$\mathbf{u}(\theta, \phi) = \left[E_n(\theta, \phi) e^{j \mathbf{k}(\theta, \phi) \cdot \mathbf{p}_n} \right]_{n=1}^4 \in \mathbb{C}^4, \quad (3)$$

and the four patterns are *distinct*. That distinctness is not a second-order correction here; it is what breaks the degeneracy discussed in Section III-A and lifts the identifiable jammer count from two to five.

A scene of J mutually incoherent jammers with received powers P_j and directions (θ_j, ϕ_j) , in spatially white noise of power σ_n^2 per element, produces the spatial covariance

$$\mathbf{R} = \sum_{j=1}^J P_j \mathbf{u}_j \mathbf{u}_j^\mathsf{H} + \sigma_n^2 \mathbf{I}_4, \quad \mathbf{u}_j \triangleq \mathbf{u}(\theta_j, \phi_j). \quad (4)$$

Incoherence is the physically appropriate model for independent hostile emitters and it matters, because summing the field amplitudes with a common phase instead — which describes coherent multipath, not independent jammers — changes the resulting measurement by tens of percent. The noise floor is -114 dBm, i.e. $\sigma_n^2 = 3.981 \times 10^{-15}$ W, and jammer-to-noise ratios are drawn uniformly on $[20, 50]$ dB throughout. We emphasise that (4) is written down here only to describe the physics: it is never available to the processor, and no quantity derived from it is ever supplied to the estimator.

2) Sectorisation

The visible hemisphere is partitioned twice. At the coarse level it is divided into 8 *sectors*, formed as the product of four azimuth quadrants and two elevation bands. The elevation split is placed at $\theta = 60^\circ$, which is the value that makes the two bands subtend equal solid angle over the hemisphere; it is not a round number chosen for convenience. Each sector is then divided into 8 *subsectors*, giving 64 subsectors over the hemisphere, each with a nominal centre direction $\mathbf{c}_{s,m} = (\theta_{s,m}, \phi_{s,m})$.

Two consequences of this partition are load-bearing later. First, the subsector centres are the only fixed directional landmarks the estimator is given, and the bias analysis shows the estimator's errors pointing toward them. Second, $\theta = 60^\circ$ is simultaneously the equal-solid-angle elevation split and a sector boundary, so any statistic binned by elevation shows structure at 60° that is partly geometric and partly a boundary effect; this caveat travels with every elevation-resolved result in this paper.

3) The Probes Are Nulling Beams

Each sector s carries a fixed codebook of $M = 8$ complex weight vectors $\{\mathbf{w}_{s,m}\}_{m=1}^8$, $\mathbf{w}_{s,m} \in \mathbb{C}^4$, one per subsector. The design intent of $\mathbf{w}_{s,m}$ is to place a deep null on the centre $\mathbf{c}_{s,m}$ of its own subsector. Probes are thus *notches, not beams*, and the entire read-out logic of the system is inverted relative to the familiar beam-scanning picture: a jammer near $\mathbf{c}_{s,m}$ drives probe m toward its minimum, and the occupied direction is announced by the *smallest* entry of the measured power vector.

The convention by which a weight is applied to the array must be stated explicitly, because it is not a matter of taste. Throughout this work the probe output for an incident field $\mathbf{u}$ is

$$s_m = \mathbf{w}_{s,m}^\top \mathbf{u}, \quad (5)$$

with the *ordinary transpose* and no conjugation. Replacing (5) by $\mathbf{w}^H \mathbf{u}$ does not perturb the null, it *relocates* it to a different direction — in general one with no counterpart on the visible hemisphere — and a codebook applied with the wrong convention exhibits almost no on-centre contrast at all. The convention in (5) is used identically by the data generator, by the bound computation and by the estimator, so that no result in this paper can be an artifact of a mismatch between them.

With (5) in force, the mean power returned by probe m is

$$\bar{p}_m = \mathbb{E}[|s_m|^2] = \sum_{j=1}^J P_j |\mathbf{w}_{s,m}^\top \mathbf{u}_j|^2 + \sigma_n^2 \|\mathbf{w}_{s,m}\|^2. \quad (6)$$

Writing $\mathbf{v}_{s,m} \triangleq \overline{\mathbf{w}_{s,m}}$ puts (6) in the compact quadratic form

$$\bar{p}_m = \mathbf{v}_{s,m}^H \mathbf{R} \mathbf{v}_{s,m} = \langle \mathbf{R}, \mathbf{v}_{s,m} \mathbf{v}_{s,m}^H \rangle, \quad (7)$$

which is the form used in Section III-A to derive the observability ceiling.

Two measurements confirm that the codebook behaves as designed. Over all 64 probes the on-centre null is 90.49 dB deep at the median and 88.36 dB at the tenth percentile, so the notch is deep and uniformly so. And a bare $\arg \min$ over the eight readings of a sector recovers the occupied subsector exactly in 70.0% of single-jammer scenes, and places it among the two deepest probes in 94.5%, against chance rates of 12.5% and 25.0%. The minimum of the probe vector is therefore a genuine, if coarse, direction estimator on its own. The estimator of this paper can be read as a learned and refined version of that $\arg \min$, and the stage-by-stage decomposition of Section V-E substantiates that reading by direct measurement rather than by analogy.

4) Measurement Statistics

The processor does not observe $\bar{p}_m$. Two distinct impairments separate the observation from the mean, and they scale differently. Finite integration leaves a *multiplicative* fluctuation on each reading, while the power detector has a *relative accuracy* floor that is additive and tied to the overall level of the scan rather than to the individual reading. The observed vector is therefore

$$p_m = \bar{p}_m \left(1 + \frac{\zeta_m}{\sqrt{K}} \right) + \varepsilon_m, \quad (8)$$

$$\zeta_m \sim \mathcal{N}(0, 1), \quad \varepsilon_m \sim \mathcal{N}(0, \alpha \bar{p}^2), \quad \bar{p} \triangleq \frac{1}{M} \sum_{m=1}^M \bar{p}_m, \quad (9)$$

with K the number of integrated snapshots and α the detector's relative-accuracy variance. We use $\alpha = 10^{-3.5} = 3.162 \times 10^{-4}$ and $K = 10^6$. Readings are floored at a small fraction of $\bar{p}$, since a physical detector cannot report negative power and the additive term can otherwise drive a deep notch below zero.

The value of K is not a free convenience. The ratio of the two variances in (8)–(9) is $(K\alpha)^{-1}(\bar{p}_m/\bar{p})^2$, so which term dominates is decided by the *probe dynamic range* and not by $K\alpha \gg 1$; a notch codebook deliberately manufactures a large dynamic range, so this is not a corner case. Section III-C quantifies the consequence and is the reason $K = 10^6$ rather than a more usual figure. At an illustrative 20 MHz sampling rate, $K = 10^6$ is 50 ms of integration per probe, or 0.40 s for a complete eight-probe scan — which is the real latency budget of the system, and it is set by the measurement, not by the estimator.

Finally, one query is issued per jammer, using that jammer's own sector codebook, but every query sees the *full* covariance (4) and therefore the leakage from all other jammers. The measured probe contrast — the spread between the deepest and the mean reading — is 20.3 dB at the median with a single jammer, falling to 6.9 dB with two and 4.8 dB with three. Multi-jammer degradation thus begins in the measurement itself, before any estimator is involved.

5) Problem Statement

Let $\mathbf{x} = (\{\theta_j, \phi_j, P_j\}_{j=1}^J, \sigma_n^2)$ denote the scene. The estimation problem is: given the observed power vector $\mathbf{p} \in \mathbb{R}_+^8$ from a sector- s query and the sector index s , recover the directions $\{(\theta_j, \phi_j)\}$. The jammer count $J \leq 3$ is taken as known, and the sector index is taken as given; acquiring the coarse sector is a separate detection problem and is outside the scope of this paper. Nothing else is supplied — in particular, no phase, no snapshots, no covariance, and no absolute power reference beyond what survives the preprocessing of the power vector itself.

This is a $(3J + 1)$ -parameter inverse problem read through $M = 8$ non-negative scalars, and whether it is solvable at all is a question settled before any estimator is chosen. Section III shows that the quadratic form (7) caps the number of accessible information dimensions at 16 irrespective of how many probes are read, and derives from that cap both the identifiability limit

on J and the per-scene accuracy bound against which every result in this paper is measured.

Accuracy metric. Because the two angles are not commensurate, error is reported as the great-circle (arc) angle between the true and estimated directions,

$$e = \arccos(\sin \theta \sin \hat{\theta} \cos(\phi - \hat{\phi}) + \cos \theta \cos \hat{\theta}), \quad (10)$$

which is invariant to the coordinate convention and does not overweight azimuth near zenith. For $J > 1$ the estimates are matched to the truths by the permutation minimising total arc error, so that a swapped pair is scored as a swap rather than as two large errors.

From directions to nulls. The estimator is not the end of the chain, and an angular error is only as interesting as the suppression it costs. Given the J estimated directions we form $\mathbf{A} = [\mathbf{u}(\hat{\theta}_1, \hat{\phi}_1), \dots, \mathbf{u}(\hat{\theta}_J, \hat{\phi}_J)]$ and the projector onto the orthogonal complement of their span, yielding the combining weight

$$\mathbf{P} = \mathbf{I}_4 - \mathbf{A}\mathbf{A}^+, \quad \mathbf{w}_* = \frac{\mathbf{P}\mathbf{c}}{\mathbf{c}^H\mathbf{P}\mathbf{c}}, \quad (11)$$

with $\mathbf{c} = \mathbf{1}$ for blind power inversion [3]. The Moore–Penrose pseudoinverse rather than an explicit inverse is essential: as two estimated directions approach one another $\mathbf{A}$ becomes rank deficient, and the pseudoinverse degrades gracefully where an inverse would return a meaningless weight. Note that (11) consumes the *estimated* directions. Supplied the true ones it nulls exactly and reports infinite suppression, which measures nothing; the whole question is whether the direction stage is accurate enough to place a useful null.

Suppression is scored by the incident-referenced jammer suppression ratio

$$\text{JSR}_j = 10 \log_{10} \frac{\|\mathbf{u}_j\|^2}{N |\mathbf{w}_*^H \mathbf{u}_j|^2}, \quad (12)$$

in which P_j cancels, so (12) characterises the spatial filter alone while the filter still depends on the scene through the estimated angles. It is a re-referenced null depth, and the re-referencing is the point: a raw null depth answers “how much better is the satellite than the jammer”, whereas (12) answers “how much of the jammer did the array actually remove”. Reported values are clipped to a 50 dB ceiling, which represents the finite dynamic range over which a null depth is physically meaningful rather than a numerical artifact of an exact projection.

What is not claimed. A per-scene Cramér–Rao bound for (8)–(9) is used throughout as a yardstick, but the estimator of this paper is biased, and on part of the hemisphere its error falls below that unbiased bound. We treat this as a diagnostic to be explained rather than a result to be advertised, and at no point in this paper do we claim to beat the Cramér–Rao bound.

II. LITERATURE REVIEW

Work on protecting satellite navigation from interference divides into three families, separated by what each one does with the corrupted measurement. The first *replaces* it, fusing GNSS with dead-reckoning or exteroceptive sensors so that a position solution survives the outage. The second *filters it in*

frequency, excising the interferer’s spectral occupancy before correlation. The third *filters it in angle*, adapting an antenna pattern so that energy arriving from the interferer’s direction is rejected at the aperture itself. Only the third rejects a wideband, in-band emitter without discarding the satellite signal along with it, and it is the family to which this paper belongs. Within it the decisive split is architectural — whether the combining happens after digitisation or before it — and that split is what determines the measurement an estimator is allowed to see. We review the three families in turn, then two further bodies of work that the present problem needs and that the GNSS anti-jamming literature has largely left untouched: the theory of what can be recovered from intensity-only measurements, and learning-based direction finding.

A. Sensor Fusion

The most widely deployed response to GNSS denial is not to clean the signal but to outlive its absence. Tightly and loosely coupled GNSS/INS integration, built on Kalman filtering and its nonlinear descendants, propagates an inertial solution through the outage and reconverges when satellites return [4], [5]; error-state and map-matching variants exploit route constraints where the platform is track-bound [6]; factor-graph formulations replace the filter with a batch estimator that tolerates intermittent and heavy-tailed measurements far more gracefully in urban settings [7]. Complementary exteroceptive sensing — cameras, lidar, and multi-platform cooperative ranging — extends the same idea to longer outages [8]–[11], and the resulting architectures now underpin much of the autonomy stack [12].

Fusion is indispensable and orthogonal to the present work, but it does not address the problem this paper poses. Inertial drift grows without bound, so the approach buys time rather than immunity; the auxiliary sensors frequently degrade in the same conditions that degrade GNSS; and the additional sensors and computation are themselves a size, weight and power burden on exactly the platforms least able to carry it. Fusion also never localises the interferer, so it cannot inform any subsequent spatial response. Continuity requirements in safety-critical transport make the residual exposure explicit [13], [14]. We add no inertial or exteroceptive sensor in this work, and treat fusion as a downstream consumer of whatever the antenna can deliver.

B. Spectral and Out-of-Band Mitigation

Where the interferer occupies a small and identifiable part of the band, it can be removed before correlation. Adaptive notch filtering and high-resolution spectral estimation excise continuous-wave and slowly swept tones with modest loss to the despread signal [15]; multi-resolution and wavelet decompositions localise narrowband bursts jointly in time and frequency and suppress them with fewer artefacts than a fixed notch [16]; and in array receivers, space-time processing extends cancellation to wideband interference by combining spatial and temporal degrees of freedom [17].

The limitation is structural rather than incidental: a purely spectral method has no spatial selectivity. Against a wideband

emitter whose occupancy covers the GNSS band, or against a fast-swept chirp whose instantaneous frequency never settles long enough for the filter to track, excision removes the satellite signal along with the jammer. Comparative assessments of single-antenna against array front ends make the crossover concrete: beyond a threshold in bandwidth and power, spatial rejection is the only remaining option [18]. That threshold is where this paper starts.

C. Spatial Filtering I: Digital Open-Loop Methods

Given per-element digital access, interference rejection is a mature science. The sample covariance matrix is formed directly from synchronous snapshots, and minimum-variance combining places nulls on interferers while maintaining whatever constraint the designer imposes on the look direction [1], [2]. Directions themselves are recovered by subspace methods that exploit the eigenstructure of the same covariance — MUSIC via the noise subspace [19], ESPRIT via a rotational invariance that removes the search entirely [20] — and the accompanying theory is unusually complete: the asymptotic performance of these estimators and their relationship to the Cramér–Rao bound have been characterised in closed form [21], with the broader parametric programme surveyed in [22]. In GNSS specifically, digital beamforming CRPAs have been demonstrated on FPGA platforms and are now documented in reference treatments of array receivers [23]–[25].

Two things should be noted about this literature for present purposes. First, its cost: 4 elements demand 4 coherent down-conversion chains and 4 converters, together with the clock distribution to keep them phase-aligned, and careful design is needed even to keep the adaptation from distorting the navigation observables it is meant to protect [26]. This is the burden that motivates an analog front end. Second, and more importantly for this paper, *every* method in this family consumes the covariance matrix or the snapshots from which it is built. None of them is defined on a measurement in which phase has already been discarded. The bound analysis of Section III is built in the tradition of [21], but it must be re-derived for a measurement functional those results do not cover.

D. Spatial Filtering II: Analog Closed-Loop Methods

Moving the combining network in front of the receiver collapses the cost to a single chain and a single converter, and collapses the observation to a single non-negative scalar: the total power out of the summing node. Because no covariance is available, the classical machinery is inapplicable, and the literature turns instead to *closed-loop search* — perturb the weights, observe the scalar, iterate. The variants differ mainly in the search strategy.

- *Power inversion.* The canonical result is that minimising total output power under a norm constraint automatically nulls the dominant interferer, since in a GNSS band the jammer is the only signal above the noise floor [3]. The idea has been carried directly into GNSS receivers and analysed in detail [27]–[29], and purpose-built null-steering front ends have been reported for high-precision

positioning [30]. Implementations typically discretise the angular domain and steer nulls sequentially, selecting at each step the codebook entry that minimises received power.

- *Evolutionary search.* Genetic algorithms and related population-based optimisers handle the non-convex, quantised weight space without gradients, and have been applied to null placement and to pattern synthesis more generally [31]–[33].
- *Gradient and LMS adaptation.* Steepest-descent and least-mean-square loops converge smoothly when a usable error signal exists [34], [35]; in GNSS the difficulty is that the reference signal the LMS formulation presumes is buried below the noise floor and is therefore unavailable before despreading.

The common cost is latency. Each of these methods is an iterative search over a scalar objective, so the number of measurements scales with the resolution of the weight or angle discretisation, and — when several interferers must be nulled in sequence — multiplies with the number of interferers. Convergence is also not guaranteed to a global optimum in a multi-jammer scene, where the objective surface acquires local minima. The present paper keeps the analog front end and its scalar measurement, keeps the null-steering codebook, and removes the search: the probe responses are read once and mapped directly to directions.

E. Power-Only Measurement, Covariance Recovery, and Degrees of Freedom

The measurement studied here — a set of squared magnitudes of linear functionals of a wavefield, with all phase discarded — is structurally a phase-retrieval problem, and that literature supplies the right intuition for what is and is not recoverable. Lifting the quadratic measurement to a linear functional of a rank-constrained Hermitian matrix converts recovery into semidefinite matrix completion, with recovery guarantees expressed in terms of the number of intensity measurements relative to the dimension of the unknown matrix [36], [37]. The lesson that transfers is that intensity-only data constrains a *covariance-like* object rather than the field itself, and that the dimension of that object, not the number of measurements, is what governs identifiability. Section III-A is an instance of exactly this accounting: the probe powers are linear in the Hermitian covariance $\mathbf{R}$, so they span at most 16 real dimensions no matter how many probes are read.

How many sources such a budget supports is the subject of the degrees-of-freedom literature in array processing. Nested and coprime array designs established that the identifiable source count is set by the dimension of the accessible covariance structure — the co-array — and not by the physical element count, and that geometry can be traded for degrees of freedom [38], [39]. The counting argument in Section III is the analogue of that result for a power-only front end, and it leads to a hard ceiling: a scene of J incoherent jammers plus noise carries $3J + 1$ real parameters, so joint identifiability requires $J \leq 5$. It also explains a discrepancy with the figure usually quoted for this architecture. Assuming a common element

pattern collapses the accessible span to 9 dimensions and would cap J at 2; that collapse is a degeneracy of the idealisation, and the full-wave pattern diversity of a real mounted array lifts it. To our knowledge neither the intensity-only recovery literature nor the degrees-of-freedom literature has previously been applied to the analog CRPA measurement, and the GNSS null-steering work reviewed above does not state an identifiability limit at all.

F. Learning-Based Direction Finding

A substantial literature now learns the map from array data to direction rather than deriving it. Deep networks trained on array snapshots or covariance features match or exceed subspace estimators where the model assumptions break down — under gain and phase miscalibration, mutual coupling, or position error [40] — and hold up at signal-to-noise ratios and snapshot counts where eigenstructure methods lose the subspace altogether [41]. The same construction transfers to acoustic arrays, which suggests the benefit is a property of the estimation problem rather than of any one propagation regime [42]. A parallel line of work learns the front end itself: convolutional and fully connected networks design hybrid precoders and combiners for arrays whose RF chain count is far smaller than their element count [43], [44]. Within GNSS, machine learning has been applied mainly to interference *characterisation* rather than geolocation — classifying jammer type from spectral or time-frequency representations [45], [46], and detecting spoofing from sequential receiver observables [47].

Two threads from the machine-learning literature inform the architecture of Section IV directly. The first is attention. Treating the probe responses as an unordered token sequence and letting a multi-head attention block model their interactions is the construction of [48], and the specific pattern used here — a task token that aggregates a set of measurement tokens and carries the readout — follows the class-token design of [49]. The second is model-based deep learning: rather than learn the whole map, embed the known physics as structure and learn only the residual the model cannot express [50], a discipline closely related to imposing physical constraints on the network’s output [51]. Our geometric anchor plus zero-initialised learned offset, constrained so that the returned direction is in the upper hemisphere by construction, is an instance of that paradigm.

The gap is nevertheless sharp. Every learned direction-finding method cited above assumes per-element digital access: the network is given snapshots, a covariance matrix, or a spectrogram, and it learns a better estimator from the *same* rich input that the classical methods consume. Its input, in other words, presupposes precisely the receiver hardware that a SWaP-constrained analog CRPA cannot afford. We are not aware of prior work that estimates direction of arrival from scalar power readings taken through an analog beamforming network, which is the measurement this paper addresses.

G. Positioning of This Work

The three families above leave a specific corner of the design space empty. Digital open-loop methods deliver accuracy and

a complete supporting theory at a hardware cost the platform cannot pay. Analog closed-loop methods pay the hardware cost once and then pay repeatedly in search latency, with no stated identifiability limit and degrading behaviour as interferers multiply. Learned estimators achieve robustness, but on inputs that assume the expensive front end. This paper occupies the remaining corner: the analog front end and its scalar power measurement are retained, the iterative search is removed, and multiple simultaneous interferers are addressed in a single forward pass.

Three commitments follow, and they shape everything reported later. First, the limit is established before the estimator, so that the accounting of Section III tells the reader which failures are the antenna’s and which are the network’s. Second, the estimator is built to match the measurement rather than retrofitted to it: because the codebook entries are nulling beams, the informative probe is the one reporting the *least* power, and the readout over 8 probe responses is therefore a soft argmin composed with a refiner able to leave the convex hull of the subsector centres — 52,575 parameters in total. Third, the reporting is diagnostic rather than promotional: the headline single-jammer accuracy is a median arc error of 3.717, we decompose that error into bias and scatter instead of quoting a ratio against the Cramér–Rao bound, and we identify dynamic range, not angular separation, as the dominant multi-jammer failure mode. We assume throughout that the occupied coarse sector is supplied by an upstream acquisition stage; the cost of relaxing that assumption is discussed in Section IX.

III. IDENTIFIABILITY AND PERFORMANCE BOUNDS

Two questions must be answered before an estimator is chosen, and neither depends on which estimator is eventually chosen. The first is whether the scene is recoverable from the measurement at all, which is a matter of counting dimensions. The second is how accurately it can be recovered when it is recoverable, which is a matter of Fisher information. This section answers both for the power-only scan of Section I-A, and then verifies the answer numerically against the generator that produces the data, so that the bound used throughout the paper is an audited object rather than an assertion.

A. The Observable Ceiling

Equation (7) exhibits every probe power as a *linear* functional of $\mathbf{R}$. The Hermitian 4×4 covariance lives in a real vector space of dimension $N^2 = 16$, so the map from scene to noiseless measurement factors through $\mathbb{R}^{16}$:

$$\mathbf{x} \mapsto \mathbf{R} \mapsto \bar{\mathbf{p}}, \quad \dim_{\mathbb{R}}\{\mathbf{R}\} = 16. \quad (13)$$

The consequence is stronger than it first appears, and it is worth stating in its sharpest form. The left map is where all the angle dependence lives; the right map is linear and fixed once the codebook is fixed. Because the factorisation is through a space of dimension 16, *no* codebook — however large, however cleverly optimised — can make the family of probe readings span more than 16 real dimensions. Adding probes adds rows to a linear map whose domain is already only 16-dimensional: the extra rows reduce variance by averaging, and they cannot

introduce an information direction that was not already there. A power-only analog front end is therefore subject to a hard observability ceiling that belongs to the antenna, not to the signal processing.

Since $\mathbf{x}$ carries $3J+1$ real parameters — two angles and one power per jammer, plus the noise level — joint identifiability of the scene requires that the parameter count not exceed the accessible dimension,

$$3J+1 \leq 16 \implies J \leq 5. \quad (14)$$

At $J=3$ the scene already consumes 10 of the 16 available dimensions, which is the structural reason the three-jammer results reported later are what they are: the problem is not becoming harder to *compute*, it is running out of room to be *observed*.

One familiar special case deserves correction, because taken at face value it would forbid most of this paper. Under the common-element-pattern idealisation $E_n \equiv E$, the element patterns factor out of (3) as a common scalar and the accessible span degenerates to 9 real dimensions, which by the same count would cap the identifiable jammer count at $J \leq 2$. That collapse is a property of the *idealisation* and not of the antenna. It is exactly the distinctness of the four full-wave patterns in (3) — amplitude and phase diversity that a real mounted array possesses and an isotropic model discards — that restores the missing dimensions. The count that applies to the array studied here is 16, and (14) is the limit that governs it.

B. The Fisher Information of a Power-Only Scan

The ceiling of Section III-A says when recovery is possible; it says nothing about accuracy. For that we compute a per-scene Cramér–Rao bound directly from (8)–(9).

Collect the scene into

$$\boldsymbol{\eta} = [\theta_1, \phi_1, \ln P_1, \dots, \theta_J, \phi_J, \ln P_J, \ln \sigma_n^2]^\top \in \mathbb{R}^{3J+1}. \quad (15)$$

The powers and the noise level enter logarithmically. This is a conditioning device, not a modelling choice: the angle columns of the Jacobian are dimensionless while a linear power column carries the physical scale of $\mathbf{R}$, and mixing them produces a Fisher matrix whose disparity between blocks is numerical rather than statistical. The angle bound we ultimately report is in any case invariant to smooth reparameterisation of the nuisances, so (15) costs nothing.

Treating the M readings of a query as independent and Gaussian — $\mathbf{p} \sim \mathcal{N}(\bar{\mathbf{p}}(\boldsymbol{\eta}), \mathbf{C}(\boldsymbol{\eta}))$ with $\mathbf{C}$ diagonal, since the fluctuations in (8) are drawn independently per probe —

$$[\mathbf{C}]_{mm} = \frac{\bar{p}_m^2}{K} + \alpha \bar{p}^2, \quad (16)$$

and because both the mean and the covariance depend on $\boldsymbol{\eta}$, the Fisher information carries two terms,

$$\mathbf{F} = \underbrace{\mathbf{J}^\top \mathbf{C}^{-1} \mathbf{J}}_{\mathbf{F}_{\text{mean}}} + \underbrace{\frac{1}{2} \text{tr} \left(\mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \eta_i} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \eta_k} \right)}_{\mathbf{F}_{\text{cov}}}, \quad \mathbf{J} = \frac{\partial \bar{\mathbf{p}}}{\partial \boldsymbol{\eta}^\top}, \quad (17)$$

symmetrised as $\frac{1}{2}(\mathbf{F} + \mathbf{F}^\top)$ to remove round-off asymmetry. The angles of jammer j are the parameters of interest and

everything else is a nuisance. Partitioning $\mathbf{F}$ accordingly into the 2×2 block $\mathbf{F}_{aa}$ for (θ_j, ϕ_j) , the nuisance block $\mathbf{F}_{bb}$ and the coupling $\mathbf{F}_{ab}$, the relevant bound is the inverse of the Schur complement

$$\mathbf{S}_j = \mathbf{F}_{aa} - \mathbf{F}_{ab} \mathbf{F}_{bb}^{-1} \mathbf{F}_{ab}^\top, \quad (18)$$

which is the *marginalised* bound: it charges the estimator for not knowing the jammer powers, the noise level, or the other jammers' directions. To make the bound commensurate with the arc error (10) we combine the two angular variances with the metric of the sphere,

$$\text{CRB}_{\text{arc}}(j) = \sqrt{[\mathbf{S}_j^{-1}]_{11} + \sin^2 \theta_j [\mathbf{S}_j^{-1}]_{22}}, \quad (19)$$

the $\sin^2 \theta_j$ factor being what prevents an azimuth bound from being reported as enormous merely because the jammer is near zenith, where azimuth is geometrically meaningless. Scenes whose Fisher matrix exceeds a condition number of 10^{12} are flagged rather than inverted; over the 900 scenes used in Section III-F the number so flagged is 0, so every bound quoted in this paper is a genuine finite marginalised bound and none is a regularised substitute. This is the bound against which the estimator is measured throughout.

C. Three Departures from the Textbook Form

Equation (17) is not the bound that a standard treatment of power-only or intensity-only estimation would produce, and the three differences are all consequential rather than cosmetic.

(i) *Both noise terms are retained, and which one matters is not obvious.* It is tempting to drop the snapshot term in (16) whenever $K\alpha \gg 1$. That test is wrong. The ratio of the two contributions is

$$\frac{\bar{p}_m^2/K}{\alpha \bar{p}^2} = \frac{1}{K\alpha} \left(\frac{\bar{p}_m}{\bar{p}} \right)^2, \quad (20)$$

so dominance is decided by the *probe dynamic range* $\bar{p}_m/\bar{p}$ and not by $K\alpha$ alone — and a codebook of deep notches manufactures a large dynamic range deliberately, so the correction is structural rather than a corner case. At the earlier setting $K = 8192$ the prefactor in (20) is 0.386, so the neglected term exceeds the retained one for any probe reading more than $1.6\times$ the sector mean — routine for the bright probes of a notch scan. With $K = 10^6$ and the measured maximum $\bar{p}_m/\bar{p} = 2.91$, the worst-probe ratio falls to 0.027; $K > 5.4 \times 10^5$ is required to hold it below 0.05. This, and not convenience, is why $K = 10^6$.

(ii) *$\mathbf{F}_{\text{cov}}$ is measured, not assumed away.* Because $\mathbf{C}$ in (16) depends on the parameters through $\bar{\mathbf{p}}$, the scene is partly observable through the *fluctuation* level of the readings and not only through their means. Omitting $\mathbf{F}_{\text{cov}}$ is at least conservative: it removes a positive semidefinite term, which lowers $\mathbf{F}$ in the PSD order, and the Schur complement (18) is matrix-monotone, so the reported bound can only become looser. We nevertheless compute it and report its measured effect in Section III-D rather than arguing it is negligible.

(iii) *The probe convention is $\mathbf{w}^\top \mathbf{u}$ throughout.* The bound is differentiated through exactly the forward model of (5) and (6). Writing the more familiar $|\mathbf{w}^\text{H} \mathbf{u}|^2$ inside the Jacobian would

TABLE I: Verification of the bound implementation. Each row is an independent check against a limit that can be derived or measured without the bound code. The Monte-Carlo row is quoted at JNR = 20/35/50 dB.

Check	Value	Expect
Detector-only limit ratio	0.999	1
$\mathbf{F}_{\text{cov}}$ on/off ratio	1.000	≤ 1
Worst-probe variance ratio	0.027	$\ll 1$
Monte-Carlo ML RMSE/CRB	0.901/0.926/0.898	≈ 1
arc-CRB spread over JNR	0.000 dB	0
Weak-jammer slope (dB/dB)	-0.850	≈ -1.000

silently relocate every null, as discussed in Section 3, and would yield a bound for a different antenna than the one the data came from.

D. Verification Against the Generator

A bound is only as trustworthy as its implementation, so the computation of Section III-B was checked against independently derived limits and against brute-force estimation. Table I collects the results.

The first three rows confirm the machinery. Switching off the snapshot term reproduces the detector-only closed form to 0.999 of its value; including $\mathbf{F}_{\text{cov}}$ changes the bound by a factor of 1.000, confirming both that it is correctly signed — a ratio above unity would indicate an error, since extra information cannot loosen a bound — and that its numerical effect at these operating points is immaterial, which is a measurement rather than the usual assumption.

The fourth row is the substantive check: a maximum-likelihood fit run directly on synthetic power vectors, 800 independent trials per operating point, attains an RMSE/CRB of 0.901, 0.926 and 0.898 at JNR = 20, 35 and 50 dB. These sit a few percent *below* unity, and the honest reading of that is not that the bound is violated but that the estimator used for the check is not exactly unbiased: the optimiser is initialised at the truth, which on a non-quadratic likelihood surface leaves a mild attraction toward it, so a small sub-unity ratio is the expected outcome of this experiment. What the row establishes is that the bound is tight to within a few percent and correctly tracks JNR — which is what it is needed for.

E. What the Bound Predicts

Beyond providing a yardstick, (17) makes two qualitative predictions that can be tested, and both hold.

The bound is independent of JNR. Raising every jammer power by a common factor scales $\bar{\mathbf{p}}$ and $\bar{p}$ by that factor, so both terms of (16) scale as the *square* of it, while the Jacobian in (17) scales linearly. The factor cancels identically in $\mathbf{J}^T \mathbf{C}^{-1} \mathbf{J}$, so the angular bound should not move at all as JNR sweeps. Measured over the full 20–50 dB range the arc bound moves by 0.000 dB. This is a genuinely non-obvious property of the front end: because the detector’s accuracy floor is *relative*, adding jammer power buys no angular information. The performance-limiting resource is the probe dynamic range, not the received power.

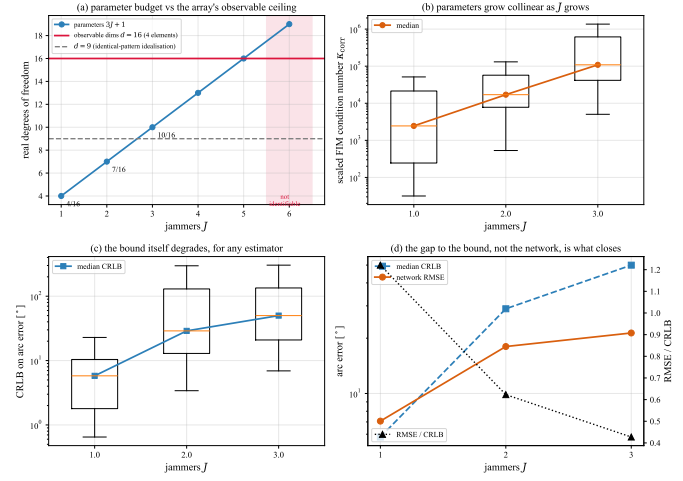


Fig. 1: Conditioning of the power-only Fisher information and the resulting bound, against the number of jammers (COND). The correlation-normalised condition number of (21) rises from 2.450×10^3 at $J = 1$ to 1.089×10^5 at $J = 3$, a factor of 44.4, and the median per-scene arc bound rises from 5.793° to 49.969° , a factor of 8.63. The estimator’s RMSE over the same span grows by only 3.02; the gap is not efficiency but bias, as Section V-C shows.

A weak jammer alongside a strong one is bounded by the power ratio. Holding the strong jammer at JNR = 45 dB and pushing a second jammer down by Δ , the weak jammer’s arc bound degrades at a measured mean rate of -0.850 dB/dB, rising from 4.633° at $\Delta = 0$ to 617.99° at $\Delta = 25$ dB — at which point the weak jammer is unlocalisable and the projection null of (11) cannot usefully be steered at it.

The measured slope is worth dwelling on, because it does not match the clean prediction. With the nuisances known, the weak jammer’s information scales as its power and the arc bound as its inverse, giving exactly -1.000 dB/dB. The measurement gives -0.850 , and the discrepancy is neither noise nor a coding error: the slope is -0.561 at $\Delta = 5$ dB and steepens monotonically to -0.992 at $\Delta = 25$ dB. Two effects explain it. First, (18) is marginalised rather than known-nuisance, and at small Δ the two jammers’ angle and power parameters are strongly coupled, so part of the weak jammer’s nominal information is spent estimating the strong one; the coupling weakens as the jammers separate in power, which is precisely why the slope steepens toward -1.000 . Second, the -1 law is an asymptotic statement, and it is recovered only once the weak jammer is faint enough for its contribution to $\bar{\mathbf{p}}$ to be a genuine perturbation. The -1.000 law is thus the limit of the measured behaviour, and the gap at moderate Δ is the cost of not knowing the rest of the scene.

F. Conditioning and the Multi-Jammer Limit

Equation (14) permits $J \leq 5$, and the array has 16 dimensions to spend. Neither fact implies that a three-jammer scene is comfortably observable, because identifiability is a statement about rank and accuracy is a statement about

TABLE II: Degrees of freedom, Fisher conditioning and the marginalised angular bound as the jammer count grows. 300 scenes per row, 900 in total; no scene was rejected by the 10^{12} guard.

J	$3J+1$	probes	used	κ_{corr}	median CRB _{arc}
1	4	8	4/16	2.450×10^3	5.793°
2	7	16	7/16	1.705×10^4	28.918°
3	10	24	10/16	1.089×10^5	49.969°

conditioning. The two part company well before the ceiling is reached, and Table II shows where.

Conditioning must be measured carefully here. The raw condition number of $\mathbf{F}$ is meaningless for this problem: the $\ln \sigma_n^2$ column of the Jacobian is of order $\sigma_n^2 \sum_m \|\mathbf{w}_m\|^2 \sim 10^{-14}$ while a jammer power column at 35 dB is of order 10^{-11} , and the quadratic form squares that disparity, so $\mathbf{F}$ reports a condition number of 5.339×10^{18} even at $J = 1$ where nothing whatever is degenerate. That number measures the choice of physical units, not the geometry. We therefore report the condition number after symmetric diagonal scaling,

$$\kappa_{\text{corr}} = \kappa(\mathbf{D}^{-1/2} \mathbf{F} \mathbf{D}^{-1/2}), \quad \mathbf{D} = \text{diag}(\mathbf{F}), \quad (21)$$

which is the standard remedy for exactly this pathology and leaves a quantity that responds to the correlation structure of the information rather than to the units of its entries.

The pattern is unambiguous. Going from one jammer to three, the number of readings available actually *triples*, from 8 to 24, because one query is issued per jammer. Despite that, the scaled conditioning degrades by $44.4\times$ and the bound itself degrades by $8.63\times$, from 5.793° to 49.969° . More measurements bought no accuracy, exactly as Section III-A said they could not: the readings are extra rows through the same 16-dimensional bottleneck, and the scene is now asking 10 questions of it instead of 4.

That the 10^{12} guard rejected 0 scenes out of 900 is the other half of the statement, and it is what makes the diagnosis specific. The three-jammer Fisher matrix is not rank deficient; the parameters remain formally identifiable. What has happened is that the information has become strongly correlated across parameters, so that recovering any one direction requires unpicking it from the others. This is a conditioning loss, and the distinction is practical rather than semantic: a conditioning loss on a full-rank problem cannot be repaired by a better read-out of the same 8 numbers, however sophisticated. It is repaired by adding independent information — more elements, hence a larger 16, or independent looks across time or frequency. A stronger estimator is not the missing ingredient, because the ingredient that is missing is not in the measurement.

This sets the expectation the estimator must be judged against, and it is a deliberately uncomfortable one: the bound says a three-jammer scene on this aperture is roughly $8.63\times$ harder than a single-jammer scene. How the trained estimator actually degrades over the same range, and why its error ratio against this bound behaves differently on well- and ill-conditioned geometries, is taken up with the measured results.

IV. ESTIMATOR ARCHITECTURE

The estimator receives exactly what the front end produces: the eight scalars of one sector query, plus the sector index. Its design is dictated by the two structural facts established above — that the probes are notches read by an $\arg \min$, and that the subsector centres are the only fixed directional landmarks available. The network is therefore built as a geometric anchor that a learned refiner corrects, rather than as a generic regressor from $\mathbb{R}^8$ to two angles.

A. The Geometric Anchor

The first branch converts the probe vector into a set of non-negative weights over the 8 subsectors of the queried sector. A small normalise–project–softmax stack maps $\mathbf{p}$ to $\mathbf{a} \in \Delta^7$, and the anchor direction is read off as the corresponding convex combination of the subsector-centre encodings,

$$\mathbf{g}_0 = \sum_{m=1}^8 a_m \mathbf{e}(\mathbf{c}_{s,m}), \quad (22)$$

$$\mathbf{e}(\theta, \phi) = [\cos \theta, \sin \theta, \cos \phi, \sin \phi]^\top,$$

the 4-component trigonometric encoding being used in place of the raw angle pair so that azimuth carries no branch cut. Equation (22) is a differentiable soft $\arg \min$: it is the learned counterpart of the bare $\arg \min$ read-out that already recovers the occupied subsector 70.0% of the time, and it inherits that read-out’s robustness rather than having to rediscover it.

A convex combination, however, cannot leave the convex hull of the 8 centres, so (22) alone can never return a direction near a sector edge. A second small head reads $\mathbf{p}$ and emits an additive offset that releases the anchor from the hull, $\mathbf{g} = \mathbf{g}_0 + \mathbf{h}(\mathbf{p})$, with $\mathbf{h}$ *zero-initialised*. The initialisation matters: at the start of training the model is exactly the soft $\arg \min$ of (22), and every departure from it is something the data paid for.

B. The Attention Refiner

The anchor is accurate to roughly a subsector. Resolving within a subsector requires reading the relative depths of all eight notches jointly, which is a comparison task, so the second branch is an attention block. The probe vector is presented as a sequence of 10 tokens of width 64: one token carrying the anchor encoding, one carrying the logarithm of the scale removed when $\mathbf{p}$ was normalised, and one per probe reading. A single 4-head self-attention layer lets every probe token be interpreted relative to the others, which is precisely the operation a notch codebook demands, since the informative quantity is a *contrast* between readings and not any reading on its own.

The separate scale token is deliberate. Normalising $\mathbf{p}$ is necessary — the absolute received power spans the 20–50 dB range and would otherwise dominate the input — but it is not harmless, because the noise term in (9) is referenced to the scan mean, so the scale carries information about how trustworthy the shape is. Supplying it as its own token preserves that information without letting it swamp the shape.

C. Hemisphere by Construction

The refiner emits a three-component correction, and the way it is applied is the one architectural choice that removes an entire class of failure. Rather than correcting θ additively and repairing the result, the correction is applied in the *logit* of $\cos \theta$,

$$\cos \hat{\theta} = \sigma(\text{logit}(\cos \theta_{\mathbf{g}}) + \Delta_{\theta}), \quad \sigma(x) = \frac{1}{1 + e^{-x}}, \quad (23)$$

so that $\cos \hat{\theta} \in (0, 1)$ and hence $\hat{\theta} \in (0^\circ, 90^\circ)$ for *any* real Δ_{θ} . The azimuth correction is a residual on the unit circle, renormalised after addition. The upper hemisphere is thus enforced by the parameterisation itself: there is no folding, no clipping and no angle wrapping anywhere in the model, and consequently no discontinuity for the optimiser to fight. The residual structure survives intact, merely expressed in logit space. That this is not a vacuous guarantee is visible in the predictions, which span 6.62° to 85.83° without a single folded output.

The whole model is 52,575 parameters. Inference is one forward pass with no iteration, no search and no matrix inversion, so the deployed latency is set by the 0.40 s probe scan of Section 4 and not by the estimator.

V. SINGLE-JAMMER ACCURACY AND THE ANATOMY OF THE ERROR

A. Accuracy

Over 10,000 single-jammer scenes with JNR drawn across the full 20–50 dB range, the arc error (10) has median 3.717° , ninetieth percentile 12.843° and RMSE 7.072° . Resolved into components, the zenith error is 3.140° at the median and 11.478° at the ninetieth percentile, the azimuth error 2.163° and 6.462° .

The gap between the median and the RMSE — roughly a factor of two — is the first thing to notice, and it is why both are quoted. The error distribution is not Gaussian but tail-dominated: most scenes are resolved to a few degrees, and a minority in which the notch carries little information are resolved very poorly and dominate any second-moment summary. Figure 2 shows the distribution directly, and an RMSE quoted alone would misrepresent it in either direction depending on the reader’s assumption.

B. Efficiency Is a Function of Geometry, Not a Number

Comparing this against the bound of Section III requires care, because the bound itself varies by more than an order of magnitude across the hemisphere: over 1,500 scenes its median is 5.020° with a tenth percentile of 0.920° and a ninetieth of 15.130° . Aggregating in quadrature gives an RMS bound of 9.464° and an overall ratio of $0.75\times$. We decline to report that number as a result. A single ratio formed across a population whose bound spans an order of magnitude is an average of two qualitatively different behaviours, and it conceals both of them.

Splitting the population at its own median bound separates them cleanly. On the well-conditioned half the network achieves 3.932° against a bound of 2.040° , a ratio of $1.93\times$; on the

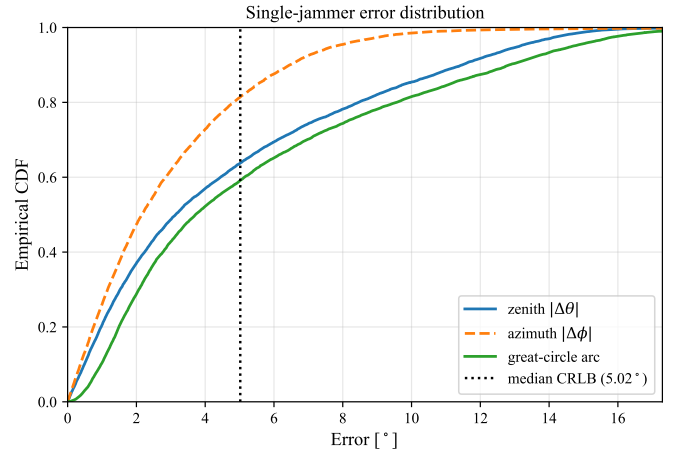


Fig. 2: Cumulative distribution of the single-jammer arc error over 10,000 scenes. The distribution is tail-dominated: the median is 3.717° but the RMSE is 7.072° , so the two summaries answer different questions and both are reported.

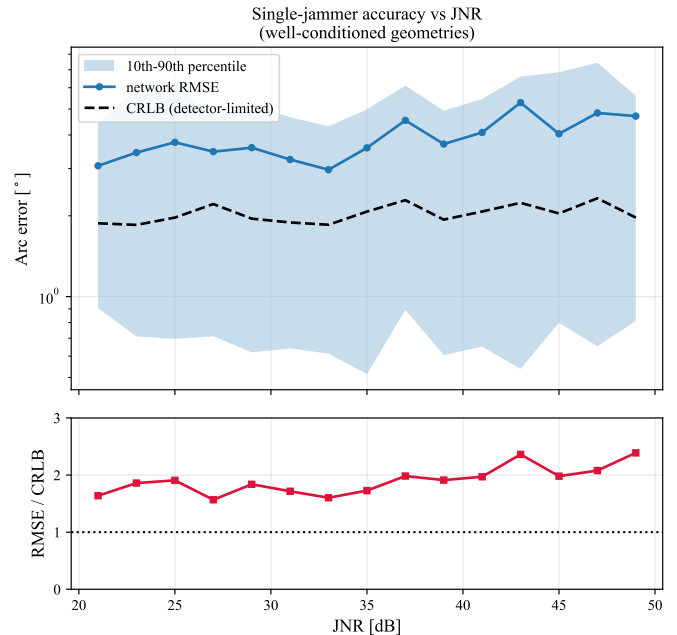


Fig. 3: Random-DoA sweep over 10,000 scenes (PH1). Accuracy is flat in jammer-to-noise ratio across the whole generator range, 7.240° RMSE in the lowest decade against 7.114° in the highest, because a notch codebook reports a ratio of powers and the ratio is insensitive to the common scale. The ratio RMSE/CRB, by contrast, is strongly geometry-dependent: 1.93 on the well-conditioned half of the scenes and 0.70 on the ill-conditioned half. The aggregate 0.75 is an average of two populations that behave oppositely and is not an efficiency.

ill-conditioned half it achieves 9.234° against 13.228° , a ratio of $0.70\times$. The estimator is thus a factor of two from efficient where the geometry is informative, and sits *below* the unbiased bound where it is not. The second number is the one that requires an explanation rather than a celebration, and

TABLE III: Fixed-DoA bias/variance decomposition, split at this experiment’s own median bound (1.985°). All quantities in degrees except the last two columns. Between the halves the bias grows $4.5\times$ while the scatter moves by 7%: the degradation is almost entirely systematic.

Half	RMSE	$\ \mathbf{b}\ $	s	bias frac.	RMSE/CRB
Well-cond.	3.812	1.217	1.540	0.415	3.19
Ill-cond.	7.775	5.470	1.642	0.894	0.68

Section V-C provides it.

One prediction of Section III-E can be checked in passing. The bound was shown to be independent of JNR; measured on this population it is 6.059° below 30 dB and 5.831° above 40 dB. The network tracks that flatness, at 7.240° and 7.114° respectively. Neither the bound nor the estimator improves with jammer power, which is the signature of a front end whose accuracy floor is relative rather than absolute.

C. Separating Bias From Variance Exactly

A ratio below unity against an unbiased bound admits only two explanations: the bound is wrong, or the estimator is biased. Section III-D has already closed off the first. Establishing the second requires separating bias from scatter, and the sweep of Section V-A cannot do it, because the direction changes on every scene, so systematic and random error are hopelessly entangled.

We therefore ran a second, differently designed experiment. The hemisphere was sampled on a grid of 21 zenith rings by 48 azimuth spokes, giving 1,008 directions, and at each direction the DoA was held *fixed* while 96 independent noise realisations were drawn at JNR = 35 dB — 96,768 queries in total. Because the truth is constant within a direction, the decomposition

$$\text{MSE}(\mathbf{c}) = \|\mathbf{b}(\mathbf{c})\|^2 + s(\mathbf{c})^2 \quad (24)$$

into a bias vector $\mathbf{b}$ and a scatter s holds *exactly* rather than approximately, and each can be measured without reference to the other.

Aggregated over the grid, the median bias magnitude is 1.994° against a median scatter of 1.575° , for a median per-direction RMSE of 2.956° and a bias fraction of 0.645. The error of this estimator is predominantly *systematic*, which already tells us that the unbiased bound is not the natural yardstick for it.

Table III is the decisive measurement. Moving from the well-conditioned to the ill-conditioned half, the bias magnitude grows from 1.217° to 5.470° — a factor of $4.5\times$ — while the scatter moves only from 1.540° to 1.642° , some 7%. The random part of the error is essentially constant across the hemisphere; everything that degrades is systematic. Correspondingly the bias fraction rises from 0.415 to 0.894, and the ratio against the bound falls from $3.19\times$ to $0.68\times$.

Of the 1,008 directions, 268 — 26.6% — have an RMSE below their own bound. Their median bias fraction is 0.884 against 0.508 for the rest. The sub-bound directions are precisely the bias-dominated ones. This is the expected behaviour

of an estimator that trades bias for variance where the geometry carries little information, and it is not superefficiency: a genuinely unbiased estimator cannot sit below the bound, and this one is demonstrably not unbiased. We therefore make no efficiency claim on the ill-conditioned half.

It is worth stating plainly that this experiment and the sweep of Section V-B are independent, with different designs, different sample counts and different median bounds — 1.985° here against 5.020° there. Their conditioning splits are therefore not the same numbers and should not be quoted as such: the fixed-DoA experiment gives $3.19\times$ and $0.68\times$, the random-DoA sweep $1.93\times$ and $0.70\times$. That two differently constructed experiments produce the same qualitative split — comfortably above the bound where the geometry is good, below it where the geometry is poor — is the point, and it is stronger evidence than either alone.

D. Where the Bias Points

A bias that merely existed would be a nuisance; this one has a direction, and the direction identifies the mechanism. For each grid point we compared the measured bias vector against the bearing from that point to its sector centre. The magnitude-weighted mean cosine is $+0.536$, and the magnitude-weighted probability that the bias points inward is 0.836 (0.712 unweighted). The bias is aimed at the sector centres — which are exactly the fixed landmarks of (22).

Figures 4 and 5 show this: the first separates the two error components direction by direction, the second places them on the hemisphere. The mechanism is the anchor’s fallback: when the notch is shallow and the probe contrast carries little information, the softmax of Section IV-A spreads its mass over several subsectors, and a spread convex combination necessarily retreats toward the interior of the hull. The learned hull offset compensates only partially. The estimator’s failure mode is therefore not noise amplification but a graceful collapse onto its own prior, which is also why the scatter barely moves while the bias grows.

This is most pronounced at high zenith angles, where the bias exceeds the scatter several-fold and the ratio against the bound is at its lowest. Two cautions attach to reading that trend off the elevation axis. First, as noted in Section 2, $\theta = 60^\circ$ is simultaneously the equal-solid-angle split and a sector boundary, so a step across it is the partition showing through rather than a property of the sky. Second, the bound itself grows steeply toward the horizon, so a falling ratio there reflects a rising denominator as much as a rising numerator.

E. Each Stage Earns Its Place, but Not Equally

Because the estimate is the sum of an analytic anchor, a learned offset applied to that anchor and a transformer correction, the output can be recomposed one stage at a time and every partial sum scored against the same truth. Over 10,000 single-jammer queries the anchor alone — the soft arg min of (22) with the learned offsets held at zero — reaches a median arc error of 10.120° and a ninetieth percentile of 20.460° . That is far better than chance, and it is a direct confirmation of the claim made in Section 3: the minimum of

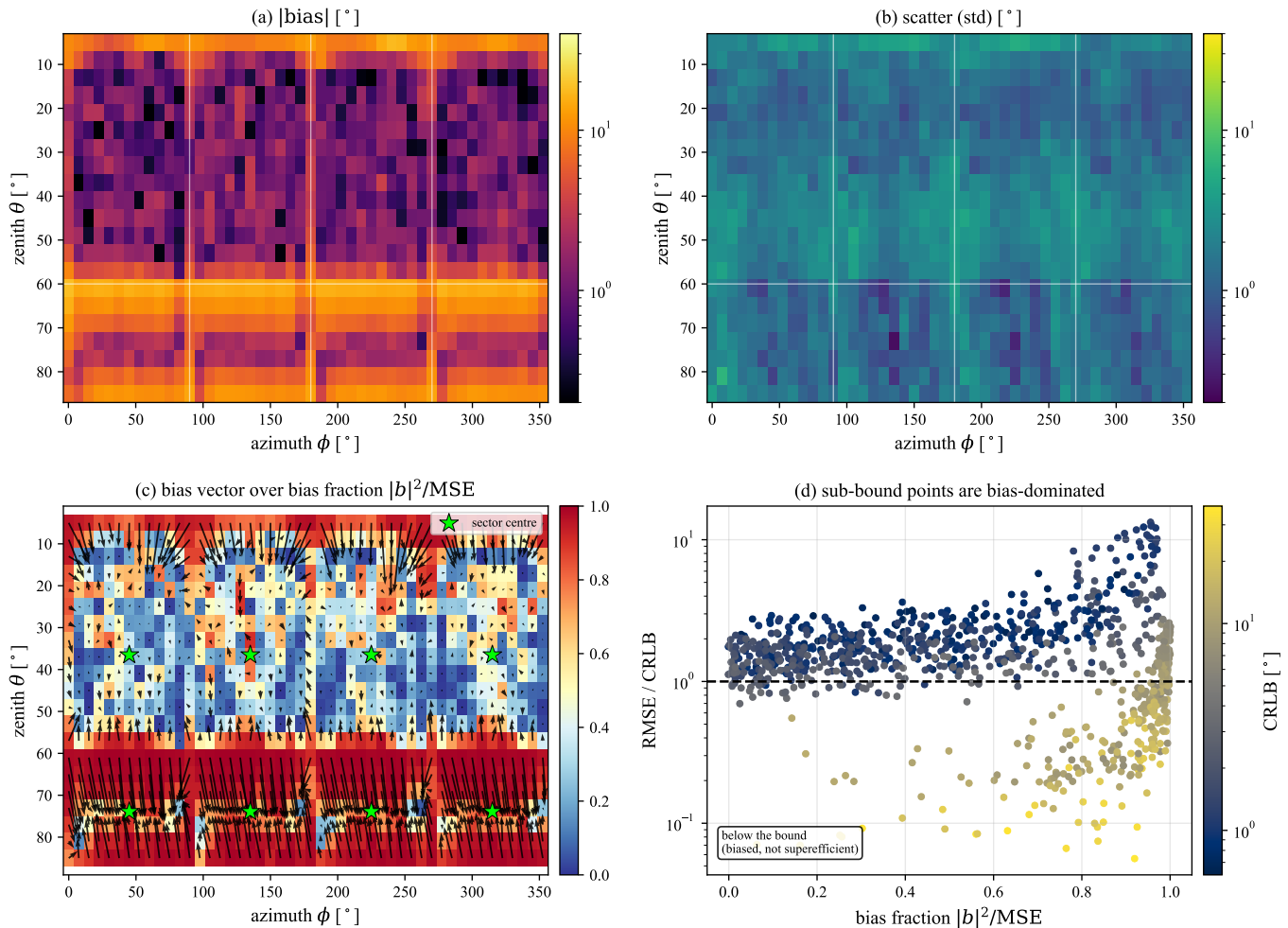


Fig. 4: The fixed-DoA experiment: the error at each of the 1,008 directions separated into its systematic and random parts according to (24). The bias grows $4.5\times$ between the conditioning halves while the scatter moves only 7%, which is what makes the sub-bound cells bias-dominated rather than efficient. The sub-panels are internal to the figure.

a notch codebook is a usable direction estimator in its own right, before any learning at all. Enabling the learned offset moves the median to 9.726° , an improvement of 0.394° , while the tail gets marginally worse, 22.054° against 20.460° . The transformer correction then brings the median to 3.717° , a further 6.009° , and the ninetieth percentile to 12.843° .

The honest reading of Fig. 6 is that two of the three stages carry the result and the third is close to idle. The anchor contributes the coarse, geometry-bound estimate that keeps the answer inside the queried sector. The refiner contributes the interpolation between subsector centres that converts a 10.120° answer into a 3.717° one, and it is where essentially all of the accuracy of the method resides. The learned offset on the anchor contributes little enough that it could be deleted for well under a degree of median error. We report that rather than suppress it for two reasons. It localises the contribution of learning in the refiner, which is the claim the architecture was designed to test. And it identifies the floor that the system falls back to when the probe vector stops being informative — a floor set by the fixed sector geometry, not by the network — which is exactly the behaviour observed in Section VII.

VI. MULTIPLE JAMMERS: DEGRADATION AND RESOLVABILITY

Section III-A predicted that this array runs out of observable dimensions quickly, and Section III-F showed the Fisher information becoming badly scaled well before the count is exhausted. This section measures what that costs. Two experiments are reported: a degradation sweep across $J \in \{1, 2, 3\}$, and a dedicated two-jammer resolvability study that asks not how large the errors are but whether the two directions are separated at all.

A. How the Error Grows with the Jammer Count

Table IV reports per-jammer arc error against J . Errors are matched to truths by the Hungarian assignment described in Section 5, so a permutation is never counted as an error, and the estimate count is J times the scene count. The median roughly triples from $J = 1$ to $J = 2$ and then grows only modestly to $J = 3$, while the ninetieth percentile grows more evenly. The shape matters more than the magnitudes: the penalty for the second jammer is much larger than the penalty for the third.

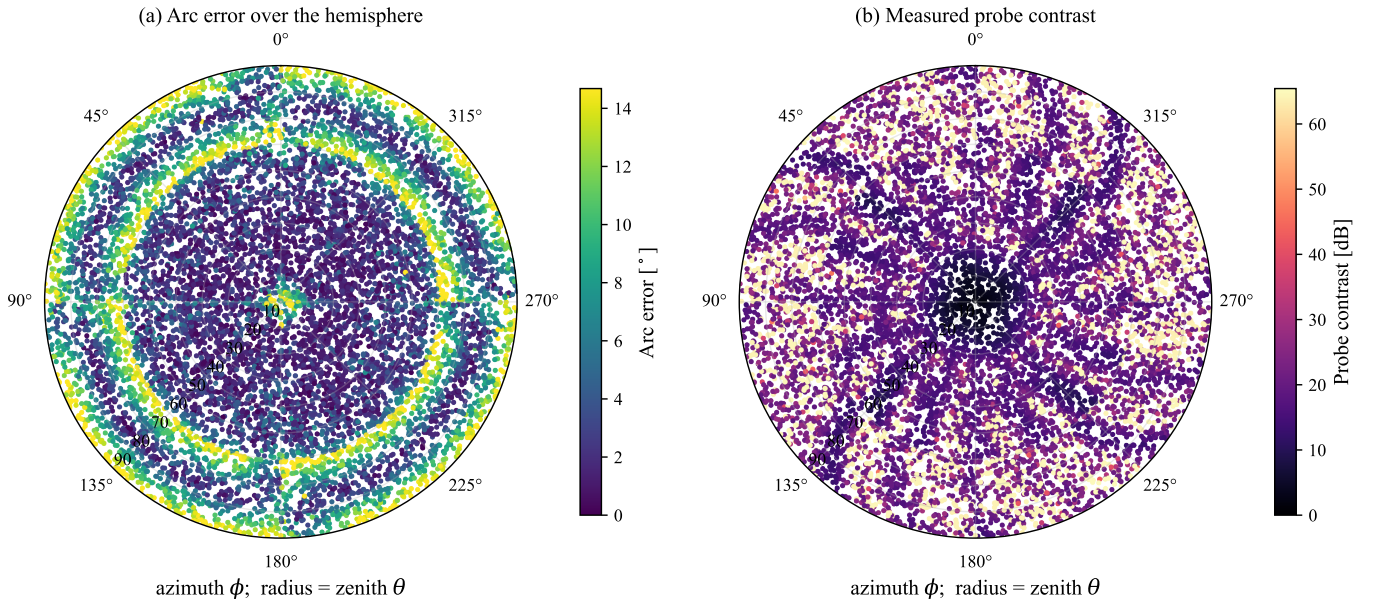


Fig. 5: The same fixed-DoA errors placed on the hemisphere. The bias vectors point inward, toward the subsector centres from which the anchor of (22) is built: the magnitude-weighted probability of pointing inward is 0.836 and the magnitude-weighted mean cosine is +0.536. Structure along $\theta = 60^\circ$ is partly geometric and partly the codebook band boundary, and the two cannot be separated with a single codebook.

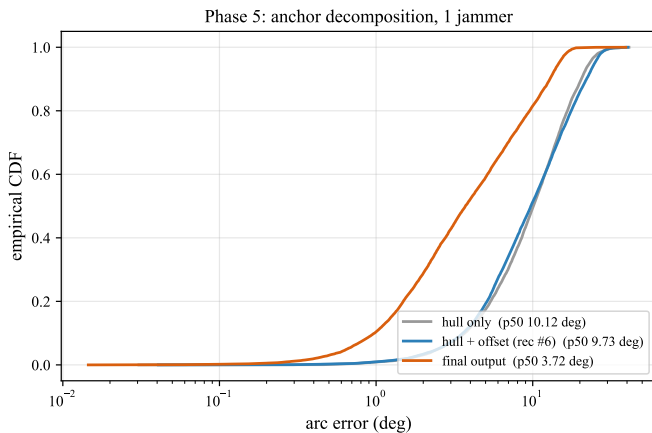


Fig. 6: Stage-by-stage recomposition of the estimator over 10,000 single-jammer queries (PH5). Anchor alone, anchor plus learned offset, and the full output are scored against the same truths. The recomposition is exact: summing the stages reproduces the forward pass to machine precision, so the comparison is an attribution and not an ablation retrained per stage.

TABLE IV: Per-jammer arc error against the number of jammers, after Hungarian matching (PH3J).

J	estimates	p_{50} ($^\circ$)	p_{90} ($^\circ$)	RMSE ($^\circ$)
1	10,000	3.717	12.843	7.072
2	20,000	11.86	29.49	18.01
3	30,000	15.26	34.27	21.35

That shape is the signature of an estimator leaning on

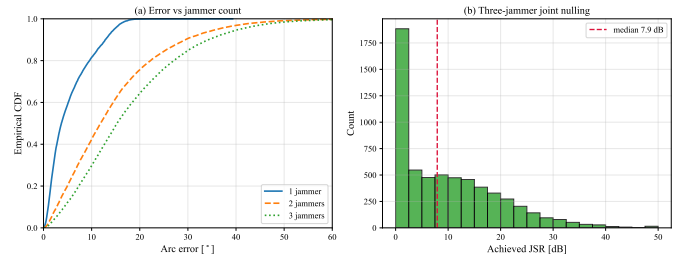


Fig. 7: Multi-jammer degradation (PH3J). Error distributions for $J \in \{1, 2, 3\}$ on common axes, together with the joint-nulling outcome at $J = 3$. With $J = 3$ nulls demanded of an 4-element array only a single spatial degree of freedom is left over, so even exact directions would purchase limited quiescent gain; the $J = 3$ figures should be read as feasibility numbers rather than as an operating point.

prior geometry. The bound behaves quite differently, as Fig. 1 showed: the median per-scene bound degrades by a factor of 8.63 between $J = 1$ and $J = 3$, while the measured RMSE degrades by only 3.02. A reader could be tempted to read the difference as increasing efficiency. It is not. The bound is the right yardstick only for an unbiased estimator, and Section V-C established by direct measurement that this estimator is biased toward subsector centres. As the scene becomes information-poor the estimate retreats toward that fixed geometry, which limits the error growth precisely because it stops tracking the data. Nothing here beats the Cramér–Rao bound; the estimator simply ceases to be the kind of estimator the bound describes.

At $J = 3$ the notion of a per-jammer error becomes strained, because the three estimates are frequently not in one-to-one

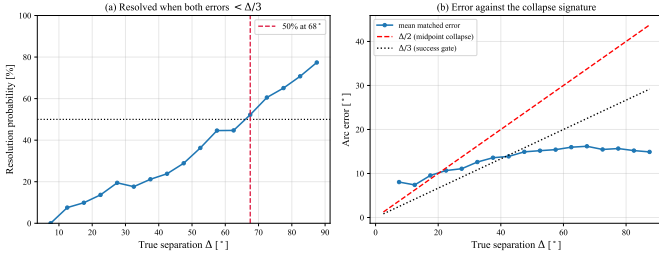


Fig. 8: Two-jammer resolvability over 10,000 scenes (PH2). Resolution rate against true angular separation, with the per-bin sample counts; the fifty-percent crossing falls near 68° . Resolution requires both estimates to lie within a third of the true separation.

correspondence with the three jammers in any useful sense. Requiring all three estimates to fall within a third of the minimum pairwise separation, the criterion used throughout this paper, succeeds in 14.0% of scenes, even though the median closest pair in those scenes is 48.9° apart — a wide separation by the standards of a 0.224λ aperture. Ten of the 16 observable dimensions are consumed at $J = 3$ and the Fisher information is badly scaled, so this is the array speaking rather than the estimator.

B. Resolving Two Jammers

A separate experiment isolates the two-jammer case, where the question is sharper: does the estimator return two distinct directions, or does it collapse onto the midpoint? Over 10,000 scenes, both directions land within a third of the true separation in 62.1% of cases. Two diagnostics confirm that this is genuine resolution. First, the Hungarian matching swaps the two labels in only 0.2% of scenes, so the pairing is essentially never ambiguous. Second, the median ratio of arc error to true separation is 0.155; collapse onto the midpoint of the pair would drive that ratio to 0.5, and a value near a sixth of that is not consistent with systematic collapse.

Resolution is governed by separation in the way spatial processing always predicts, and Fig. 8 shows the trend. In the $15\text{--}20^\circ$ separation bin only 9.9% of pairs are resolved; by the $85\text{--}90^\circ$ bin the rate is 77.4%, with the fifty-percent crossing near 68° . For an aperture of 0.224λ that crossing is unsurprising, and it is worth stating plainly that the useful two-jammer regime for this antenna is the widely separated one.

Power disparity matters less than separation, which is the more interesting result. Splitting the same scenes by the jammer-to-noise difference, pairs within 5 dB of each other resolve at 65.1% ($n = 3,073$) and pairs more than 15 dB apart at 57.4% ($n = 2,560$). The penalty for a strong jammer masking a weak one is real but modest, roughly eight points, against the nearly eightfold swing produced by separation alone. The reason is the notch codebook: because each probe suppresses its own centre, a strong jammer does not saturate the reading of the probe that is nulling it, and the weak jammer's contribution survives in that reading. This is a genuine advantage of a nulling probe

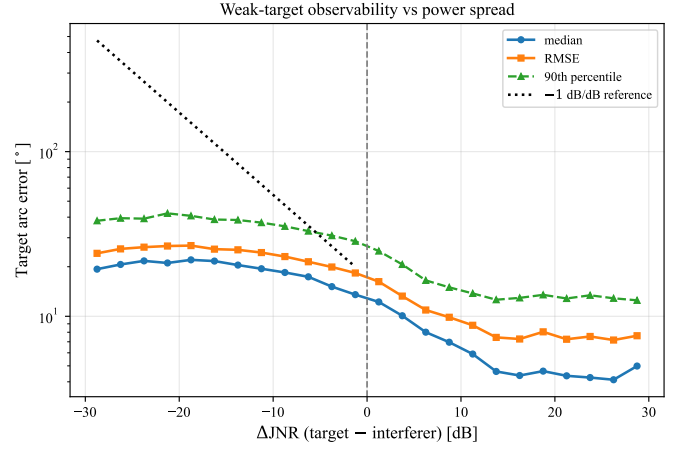


Fig. 9: Accuracy against target-to-interferer power ratio over 20,000 evaluation points (PH35). The measured trend for disadvantaged targets is $+0.051$ dB per dB, against the $+1.000$ predicted by the weak-source law of Section III-E; instead of diverging, the error saturates near 26.2° beyond about -15 dB of disparity.

set over a beam-forming one, and Section VII quantifies where it finally breaks down.

VII. DYNAMIC RANGE: THE DOMINANT FAILURE MODE

The two-jammer results above are aggregated over power disparity. Resolving that axis explicitly gives the clearest single statement of this system's limitation, and it is the one result that does not follow the theory of Section III.

Each of the 10,000 two-jammer scenes contributes two target/interferer assignments, giving 20,000 evaluation points ordered by the target's jammer-to-noise ratio minus the interferer's. The asymmetry is large. Restricting to targets at least 10 dB weaker than their interferer, the RMSE is 25.62° ; restricting to targets at least 10 dB stronger, it is 7.82° , a ratio of 3.28. Across the extreme bands the medians run from 21.10° for the most disadvantaged targets to 4.26° for the most advantaged.

The interesting part is that the estimator does not obey the law that governs the bound. Section III-E showed that the per-scene bound for a weak source degrades at very nearly -1.000 dB per dB of disparity, measured at -0.850 dB per dB over the sampled range, which is to say the bound diverges as the target is buried. The network's error does not diverge. Its measured trend over the disadvantaged bands is $+0.051$ dB per dB — flat to within the resolution of the experiment, against $+1.000$ predicted — and beyond roughly -15 dB of disparity the RMSE simply saturates at about 26.2° and stays there.

A saturating error is not a sign of strength, and it should not be read as one. The plateau value is diagnostic. The hemisphere is divided into 8 sectors, so a sector subtends a characteristic angular radius of about 28.6° , while a subsector subtends about 17.9° . The observed plateau of 26.2° sits at the sector scale, not the subsector scale, which is what one expects of an estimator that has fallen back on its coarsest prior. This is precisely the mechanism identified in Section V-E: when the probe vector

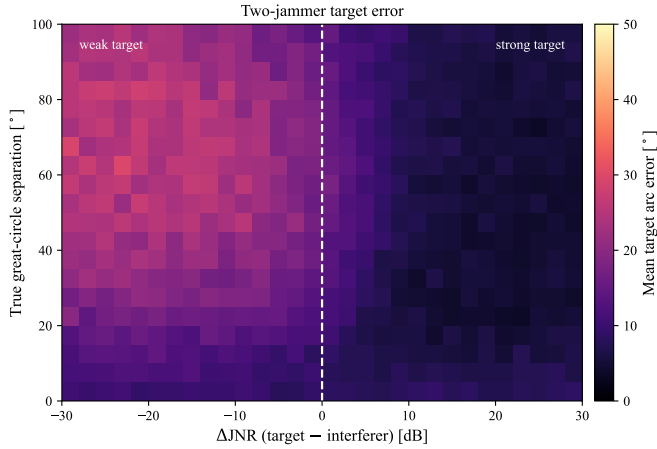


Fig. 10: Error over the joint space of the two jammers' directions and powers (PH35). The evaluation split populates only 78.0% of cells at the five-sample threshold, so the map is also drawn on a synthesised grid of 18,000 scenes retained from 1,500,000 draws, which fills every cell. With the grid complete, an empty cell means a geometry the physics does not permit rather than one the sampler happened to miss.

carries no usable information about a buried target, the output reverts to the anchor, and the anchor can be no better than the geometry it is built from. The estimator therefore fails gracefully rather than catastrophically, returning a sector-scale guess instead of a divergent one, but a sector-scale guess is not a detection. Any deployment that must find weak emitters alongside strong ones needs either a deeper codebook or a second look with the strong jammer already nulled.

One methodological point travels with Fig. 10. On the evaluation split only 78.0% of the cells of the joint direction/power map reach the five-sample threshold, which makes an empty cell ambiguous: it may be a geometry that cannot occur, or merely one the random sampler did not visit. We therefore redrew the map on a purpose-built grid, retaining 18,000 scenes from 1,500,000 draws, which populates every cell. Only then can a blank region be read as a physical statement.

VIII. FROM DIRECTION TO SUPPRESSION

Direction error is an intermediate quantity. What a receiver experiences is the jammer-to-signal ratio after nulling, and that is what this section reports. Estimated directions are turned into weights by the projection nulling of (11), and the resulting JSR of (12) is evaluated against the same full-wave patterns used to generate the measurement. The figure is clipped at a 50 dB ceiling and floored at zero, so what is reported is deliberately conservative: no scene is credited with more suppression than a real front end would deliver.

A. Single Jammer

Over 4,000 single-jammer scenes, with no scene discarded as degenerate, the median JSR is 27.71 dB, with a tenth percentile of 19.77 dB and a ninetieth of 37.71 dB. Expressed as coverage, 98.6% of scenes achieve at least 10 dB of suppression, 89.3%

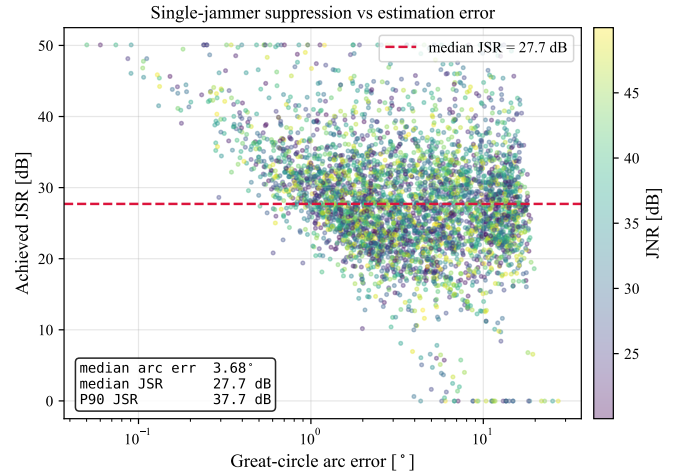


Fig. 11: Suppression against direction error over 4,000 single-jammer scenes (PH4). The relationship is strikingly flat: the median JSR falls only from 34.24 dB for scenes accurate to within 1° to 28.21 dB for scenes accurate to within 5° . A null cast by a 0.224λ aperture is broad, so the suppression tolerates a direction error of several degrees almost without penalty.

at least 20 dB, and 37.3% at least 30 dB. The tenth percentile is the number an integrator should plan against, and it is comfortably above the threshold at which a GNSS receiver recovers lock.

Fig. 11 makes the operationally important point. Conditioning on direction accuracy, the median JSR is 34.24 dB for scenes with arc error within 1° ($n = 423$), 29.17 dB within 3° ($n = 1,735$), and 28.21 dB within 5° ($n = 2,376$). A fivefold relaxation of the accuracy requirement costs about 6 dB. The explanation is the aperture: a null formed on 0.224λ is angularly broad, so the pattern null does not have to be placed precisely to place the jammer deep inside it. This is why a median direction error of 3.717° is not merely acceptable but close to sufficient, and it is the reason a modest estimator on a small array remains useful.

B. Two and Three Jammers

With two jammers the dynamic-range asymmetry of Section VII propagates straight through to suppression. Over 4,000 two-jammer scenes, evaluated once per jammer, the mean JSR is 24.20 dB for the stronger jammer and 6.54 dB for the weaker one. The weaker jammer is both harder to locate and, having been located poorly, less well nulled.

Splitting instead by geometry gives a result that reads backwards at first sight: widely separated pairs, at least 60° apart, average 14.04 dB, while closely spaced pairs under 30° average 17.76 dB. Closer jammers are suppressed *better*. The likely mechanism is that two nulls demanded of an 4-element array consume both spare degrees of freedom when the directions are far apart, and each must then be placed accurately on its own target; when the two directions are close, a single broad null covers both, and the direction errors — which are correlated in that regime — move the pair together inside a null wide enough to hold them. We offer this as

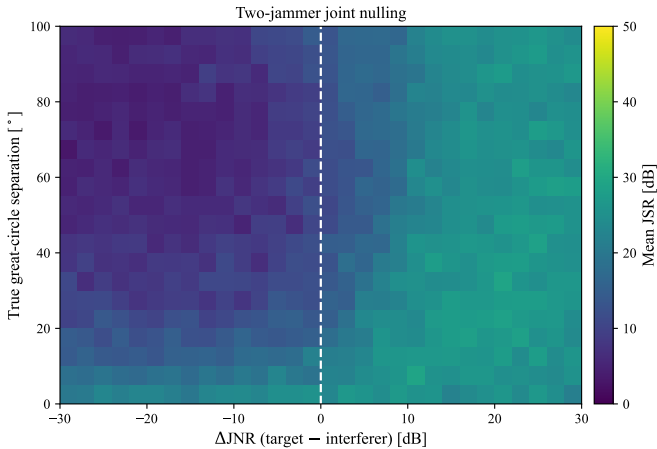


Fig. 12: Two-jammer suppression over the joint direction/power space (PH4). As in Fig. 10, the evaluation split reaches the five-sample threshold in only 61.3% of cells, so the map is redrawn on the synthesised grid, where every cell is populated and the weak/strong means are 8.69 dB and 25.18 dB.

the most plausible reading of the measurement rather than as a demonstrated mechanism; separating the two contributions would require an ablation we have not run.

Three jammers are reported for completeness and should be read as a feasibility check. Over 2,000 scenes, with no degenerate geometry encountered, the mean JSR is 9.91 dB and the median 7.93 dB, with 43.2% of scenes reaching 10 dB. The limitation is structural rather than algorithmic: casting three nulls with 4 elements leaves a single degree of freedom for everything else, so even exact directions would purchase only limited quiescent gain. Improving this number requires more elements, not a better estimator.

IX. DISCUSSION AND LIMITATIONS

The results support a narrow claim, and it is worth stating what the claim does and does not include.

The sector index is an input. Every result in this paper is conditioned on knowing which of the 8 sectors is being queried, because the codebook is per-sector. Acquisition of the sector — scanning the sectors, or fusing their probe vectors — is outside the scope of this work. A full system would either sweep all 8 sectors, multiplying the 0.40 s scan time of Section 4 accordingly, or use a coarse omnidirectional detector to select one. We consider this the most significant gap between what is measured here and a deployable receiver.

The elevation split does double duty. The band boundary at $\theta = 60^\circ$ is simultaneously the equal-solid-angle split and a codebook boundary. Every elevation-resolved result therefore mixes a geometric effect with a boundary effect, and we have flagged this wherever such a result appears. Disentangling the two requires a second codebook with the split moved, which we have not built.

The array is simulated, not measured. The element patterns are full-wave simulations of the mounted array, which captures pattern diversity and mutual coupling as modelled, but not manufacturing tolerance, mounting variation or calibration error.

The observability argument of Section III-A depends on pattern diversity being real, and that dependence should be validated on hardware before the $J \leq 5$ ceiling is relied upon in a design.

The bound is not the right yardstick, and we do not claim to beat it. The estimator is measurably biased toward subsector centres. On ill-conditioned geometries its RMSE falls below the unbiased Cramér–Rao bound, and Section V-C shows by direct measurement that those cells are bias-dominated: the bias grows by 4.5 between the conditioning halves while the scatter moves by 7%. A biased estimator is not governed by the unbiased bound, so the sub-bound cells are a statement about our estimator’s class, not an achievement. No claim of bound violation is made anywhere in this paper.

Dynamic range is the dominant weakness. The plateau of Section VII means a target buried more than about -15 dB below its interferer is answered at sector scale, roughly 28.6° , rather than tracked. Graceful degradation is preferable to divergence, but it is not detection, and this is the first thing we would attack in future work.

The jammer count is assumed known. Results are reported at fixed J . Estimating J from the probe vector is a natural extension and is made plausible by the observability accounting of Section III-A, but it is not attempted here.

X. CONCLUSION

An analog CRPA that reports only 8 scalar powers per sector query carries enough information to steer nulls onto multiple jammers in real time. We made that statement precise in three steps. First, the observable content of any power-only scan of an 4-element array is bounded by the 16 real dimensions of the spatial covariance, which caps joint identifiability at $J \leq 5$ and shows the widely quoted limit of 2 jammers to be an artifact of the common-element-pattern idealisation rather than a property of a real mounted array. Second, we derived the Fisher information of the measurement in the form the hardware actually delivers it and verified it against the generator, which supplies a per-scene bound and, more usefully, a per-scene conditioning number that predicts where the problem is hard. Third, we built a 52,575-parameter estimator whose output lies in the upper hemisphere by construction and which reads the notch structure of the codebook directly, reaching a median arc error of 3.717° on single jammers and delivering a median 27.71 dB of suppression after projection nulling, with 89.3% of scenes above 20 dB.

The diagnostic results are as much of the contribution as the accuracy figures. Efficiency relative to the bound is not a number for this problem but a function of geometry, 1.93 on well-conditioned scenes and 0.70 on ill-conditioned ones; a fixed-direction experiment shows the sub-bound cells to be bias-dominated, with the bias pointing inward toward the subsector centres the anchor is built from; and the dynamic-range sweep shows the estimator abandoning the weak-source law entirely in favour of a sector-scale fallback. Each of these is a property of an estimator that has been given a strong geometric prior, and each of them tells a designer something the aggregate accuracy figures conceal. The immediate next steps follow from the limitations: sector acquisition, a dynamic-range-aware

codebook or a two-pass scan, and validation of the pattern-diversity argument on measured hardware.

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